
BRIDGING THE NUTRITIONAL LITERACY GAP: A CONTENT-DEPENDENT FRAMEWORK FOR AUTOMATED FOOD SAFETY ASSESSMENT IN THE INDIAN MARKET

Archit Srivastava^{*1}, Arham Khan^{*2}, Divyansh Singh^{*3},
Amitabh Gupta^{*4}, Mrs. Aliza Raza Rizvi^{*5}

^{*1,2,3,4}Department of Computer Science & Engineering Babu Banarasi Das Northern India Institute of
Technology (BBDNIIT) Lucknow, Uttar Pradesh, India.

^{*5}Guide/Supervisor, Department of Computer Science & Engineering Babu Banarasi Das Northern
India Institute of Technology (BBDNIIT) Lucknow, Uttar Pradesh, India.

DOI: <https://www.doi.org/10.56726/IRJMETS87732>

ABSTRACT

The Indian food ecosystem is characterized by a dichotomy: a rapidly expanding market of ultra-processed foods (UPFs) coexisting with rising rates of non-communicable diseases (NCDs) such as diabetes and hypertension. While regulatory frameworks mandate nutritional transparency, widespread "nutritional illiteracy" remains a critical barrier. Consumers frequently struggle to interpret complex chemical nomenclature or read microscopic fonts on packaging, often prioritizing brand recognition over nutritional quality. Existing digital interventions predominantly rely on barcode scanning (UPC/EAN) linked to global databases. This "Database-Lookup" model systematically excludes the vast sector of regional, unbranded, and local food products ("Namkeen," bakery items) prevalent in the Indian market, effectively rendering these tools useless for a significant demographic.

This paper proposes a novel Content-Dependent Ingredient Analysis Framework. Unlike traditional systems, our solution employs Cloud-Native Optical Character Recognition (OCR) via Microsoft Azure AI to extract ingredient data directly from product packaging, independent of database indexing. The system integrates a robust Heuristic Scoring Engine that evaluates risks associated with hidden additives, Non-Nutritive Sweeteners (NNS), and potential adulterants. Experimental validation demonstrates that this architecture overcomes the geometric distortion limitations of client-side OCR on cylindrical packaging (96.4% accuracy vs. 42.1%) and achieves 100% analysis coverage for local Indian snacks where barcode apps fail.

Keywords: Digital Health, Optical Character Recognition (OCR), Food Safety, Consumer Informatics, Azure AI, Heuristic Analysis, Ultra-Processed Foods.

I. INTRODUCTION

A. The Public Health Context

The consumption of Ultra-Processed Foods (UPFs) in Lower-Middle-Income Countries (LMICs) like India is on a precipitous rise. These products often replace nutrient-dense whole foods, contributing to a "Health Halo" effect where consumers mistakenly believe a product is healthy due to misleading marketing labels like "Low Fat" or "Natural." However, the reality is often obscured by chemical complexity. Ingredients such as Non-Nutritive Sweeteners (NNS) (e.g., Aspartame, Sucralose) are increasingly used to replace added sugars. While this reduces caloric content, observational studies link habitual NNS consumption to metabolic disruptions and an increased preference for sweet tastes, perpetuating a cycle of poor dietary habits.

B. The Behavioral and Technical Barrier

Despite the availability of nutritional labels, Singla et al. (2010) report that Indian consumers face significant hurdles in utilizing them. The primary barriers include:

1. Legibility: Font sizes are often too small for comfortable reading.

2. Terminology: Technical names for common additives (e.g., "Tertiary Butylhydroquinone" for a preservative) are unintelligible to the layperson.

3. Behavioral Bias: Consumers prioritize brand reputation over actual nutritional content.

Furthermore, current technological interventions like MyFitnessPal or Yuka operate on a rigid Database-Dependent Model. They require a product's barcode to be pre-indexed in a central server. This model fails catastrophically in the Indian context, where a significant portion of the food supply comes from the unorganized sector (local bakeries, regional snack manufacturers) that do not maintain digital UPC records.

C. Proposed Solution

This research introduces the "Ingredient Analyzer," a web-based intervention tool that shifts the paradigm from identifying the product to reading the product. By leveraging deep learning-based Computer Vision to read text directly from the package, the system democratizes access to food safety analysis, ensuring that even a locally manufactured packet of "Rusks" can be analyzed for hidden sugars and preservatives with the same efficacy as a global brand.

II. LITERATURE SURVEY

A. Digital Health Implementation in LMICs

Gudi et al. (2023) emphasize that digital health interventions in countries like India face unique infrastructure challenges, including low digital literacy and variable network connectivity. While the National Digital Health Mission (NDHM) aims to digitize health records, there is a gap in preventative, consumer-facing tools. Successful interventions must be lightweight, intuitive, and resilient to intermittent connectivity—principles that guided our web-based (PWA) architectural choice over a heavy native application.

B. Risks of Hidden Additives and Adulteration

The necessity for rigorous ingredient analysis is underscored by the risks of adulteration. Mahima et al. highlight that traditional herbal products, often consumed for health benefits, frequently suffer from substitution errors or heavy metal contamination (arsenic, mercury). Similarly, Russell et al. (2023) found that while added sugar sales are decreasing in high-income nations, they are increasing in LMICs. Our system's Heuristic Engine is specifically designed to flag these high-risk categories, including "hidden" sugars and ambiguous herbal ingredients.

C. Limitations of Automated Extraction Methods

Automating the extraction of text from food packaging is a non-trivial computer vision problem.

1. Geometric Distortion: Gao et al. demonstrated that standard OCR models fail when applied to curved surfaces (e.g., beverage cans) due to character deformation. They proposed complex Rectification-CNN pipelines, which are computationally expensive for mobile devices.

2. Crowdsourcing Latency: Noronha et al. proposed "PlateMate," a system relying on crowdsourced human workers. While accurate, it incurred an average latency of 94 minutes per request, rendering it non-viable for real-time in-store decision making.

3. Client-Side OCR Failure: Our preliminary testing with Tesseract.js (a popular open-source library) revealed error rates exceeding 60% when analyzing reflective or crinkled packaging, common in the Indian snack industry.

D. Information Integrity

Denniss et al. conducted a systematic review revealing that 48.9% of online nutrition-related information is of low accuracy. This validates our approach of analyzing the physical label—the legally binding source of truth—rather than relying on potentially outdated or user-generated database entries.

III. SYSTEM ARCHITECTURE

The Ingredient Analyzer employs a Three-Tier Decoupled Architecture designed to balance high-accuracy processing with low-latency user experience.

A. Presentation Layer (Frontend)

Developed using React.js and Tailwind CSS, the frontend serves as a lightweight interface accessible via any standard mobile browser.

- **Image Optimization:** To address the bandwidth constraints in India, we integrate the browser-image-compression algorithm. This downscales high-resolution camera captures (often 5MB+) to a target payload of <1MB before transmission, reducing API latency by approximately 60% without compromising OCR legibility.

B. Application Layer (Backend)

The core logic resides in a Node.js/Express.js environment. This layer acts as the orchestrator:

1. **API Gateway:** Securely routes image payloads to the Azure Cloud.
2. **Sanitization Pipeline:** The raw text returned by the OCR engine is "noisy," containing manufacturing dates, FSSAI license numbers, and address details. We implement a Regular Expression (Regex) filter to isolate the Ingredients: block.
3. **Fuzzy Logic Module:** To mitigate minor OCR errors (e.g., reading "Maltcdextrin" instead of "Maltodextrin"), we utilize the Levenshtein Distance algorithm. If the edit distance between an extracted token and a database entry is ≤ 2 , it is treated as a valid match.

C. Intelligence & Data Layer

1. **Cloud OCR (Azure AI Vision):** We utilize Microsoft Azure's pre-trained Transformer models (Read API v3.2). Unlike local OCR, these models are trained on "text-in-the-wild," enabling them to handle the specular reflection (glare) on plastic wrappers and the cylindrical curvature of bottles.
2. **Knowledge Base (PostgreSQL):** A relational database stores a curated library of ingredients mapped to their health impacts (e.g., Class: Preservative, Risk: High, Description: Potential Carcinogen).

IV. METHODOLOGY AND SCORING ALGORITHM

The system transforms raw text into a quantifiable "Health Score" through a deterministic, rule-based process.

A. The Penalty/Reward Heuristic

We model the health verdict as a baseline score of 50 (Neutral), modified by the presence of specific chemical attributes.

$$Score_{final} = \text{Clamp}(50 + \sum P_{tox} + \sum R_{nut}, 0, 100)$$

1. Toxicity Penalties (P_{tox}):

- **High-Risk Additives (-10):** Ingredients linked to severe health risks (e.g., Sodium Benzoate, Potassium Bromate, Partially Hydrogenated Oils).
- **Hidden Sugars (-5):** Detecting aliases like "High Fructose Corn Syrup," "Invert Sugar," "Malt Syrup."
- **Non-Nutritive Sweeteners (-5):** Detecting Aspartame, Saccharin, or Sucralose, flagging the potential metabolic risks associated with NNS.

2. Nutritional Rewards (R_{nut}):

- **Whole Foods (+5):** Ingredients like "Whole Wheat," "Oats," "Almonds," "Milk."
- **Clean Label (+2):** Natural spices or herbs without generic "flavouring" tags.

B. Verdict Classification

The numerical score is mapped to a semantic label to aid consumer decision-making:

- **Clean (80-100):** Predominantly natural ingredients.

- Moderate (50-79): Contains processed elements but within acceptable limits.
- Hazard (0-49): Contains high-risk additives or excessive hidden sugars.

V. RESULTS AND DISCUSSION

A. Coverage Analysis: The "Local Product" Gap

To evaluate the system's inclusivity, we tested 20 food items sourced from a Tier-2 city market (Lucknow), categorized by market presence.

Product Category	Sample Size	MyFitnessPal (Database Lookup)	Ingredient Analyzer (Proposed)
Global Brands (e.g., Lays, Pepsi)	5	100% Success	100% Success
National Brands (e.g., Haldiram)	5	100% Success	100% Success
Local/Regional (e.g., Bakery Rusk)	5	0% Success (Not Found)	100% Success
Unbranded/Street (e.g., Chips)	5	0% Success (Not Found)	80% Success

Discussion: The database-dependent model failed entirely for 50% of the test set (Local/Unbranded items). The Ingredient Analyzer successfully processed these items, proving its utility for the unorganized food sector.

B. OCR Efficacy on Geometric Surfaces

We benchmarked our Azure-based Cloud OCR against the standard client-side library (Tesseract.js).

Surface Topology	Tesseract.js Accuracy	Azure AI Accuracy	Improvement
Planar (Biscuits)	82.5%	99.1%	+16.6%
Cylindrical (Bottles)	42.1%	96.4%	+54.3%
Reflective (Plastic Packs)	35.8%	94.2%	+58.4%

Discussion: The drastic failure of local OCR on cylindrical surfaces confirms the findings of Gao et al. regarding character deformation. By offloading this task to the Cloud, we achieve near-perfect accuracy, which is critical when a single missed letter (e.g., confusing "Oat" with "Fat") could alter the health verdict.

VI. CONCLUSION

This research validates that a Content-Dependent, cloud-native approach is superior to database-lookup models for the diverse Indian food market. By successfully analyzing local and regional products that major apps ignore, the Ingredient Analyzer bridges a critical gap in digital health equity. It addresses the challenge of "Nutritional Illiteracy" by translating complex chemical data into actionable safety ratings.

VII. FUTURE SCOPE

1. Personalized Medical Profiling (Level 3): Integrating user health records to provide context-aware alerts (e.g., flagging "Sodium" specifically for Hypertensive users).
2. Drug-Food Interaction Check (Level 4): Implementing a logic layer to cross-reference ingredients with a user's medication list, preventing adverse interactions (e.g., Grapefruit juice with Statins).

VIII. REFERENCES

- [1] Russell, C., Baker, P., et al. "Global trends in added sugars and non-nutritive sweetener use in the packaged food supply." Public Health Nutrition, vol. 26, no. 5, 2023.

-
- [2] Gudi, N., Yadav, U.N., et al. "Context and Opportunity in Digital Health: Implementation in Low- and Middle-Income Contexts." *BMJ Innovations*, vol. 9, pp. 19–22, 2023.
 - [3] Singla, M. "Usage and understanding of food and nutritional labels among Indian consumers." *British Food Journal*, vol. 112, no. 1, pp. 83–92, 2010.
 - [4] Gao, F., Li, S., et al. "Text Spotting for Curved Metal Surface: Clustering, Fitting, and Rectifying." *IEEE Access*, 2022.
 - [5] Noronha, J., Hysen, E., et al. "PlateMate: Crowdsourcing Nutrition Analysis from Food Photographs." *UIST Proceedings, ACM*, 2011.
 - [6] Mahima, K., et al. "Advancements and future prospective of DNA barcodes in the herbal drug industry." *Frontiers in Pharmacology*, vol. 13, 2022.
 - [7] Denniss, E., et al. "Quality and accuracy of online nutrition-related information: a systematic review." *Public Health Nutrition*, vol. 26, no. 7, pp. 1345–1357.
 - [8] Microsoft. "Azure AI Vision Documentation – Read API v3.2." Microsoft Learn, 2024.